



**THE INSTITUTION OF ELECTRONICS AND
TELECOMMUNICATION ENGINEERS
CHENNAI CENTRE**

**ELITECCOM 2025
PAPER PRESENTATION**

ORGANIZATION NAME:
Sri Venkateswara College of Engineering

**PAPER NAME: AI POWERED SMART BAND FOR
HEALTH MONITORING**

Paper ID: PP25

TEAM MEMBERS



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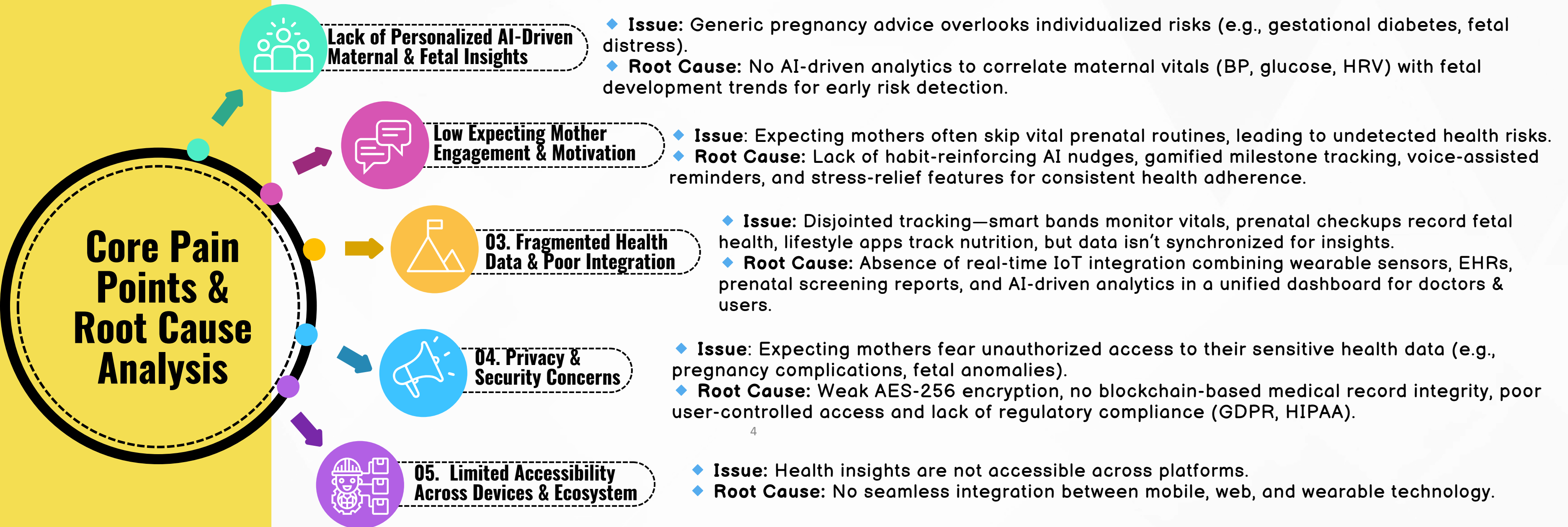


PARTHIBAN V
MEMBER



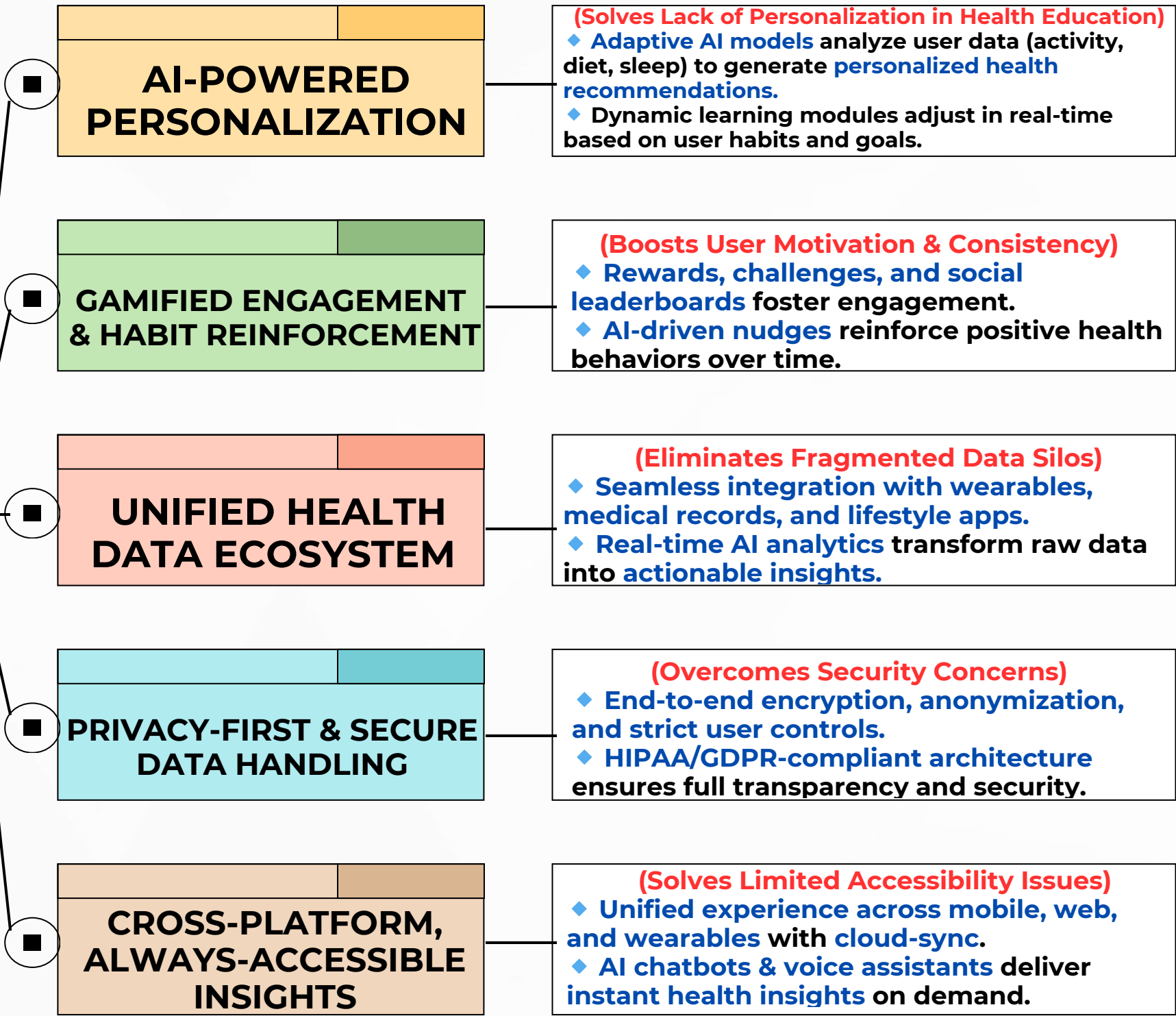
PRANAY V
MEMBER

PROBLEM STATEMENT

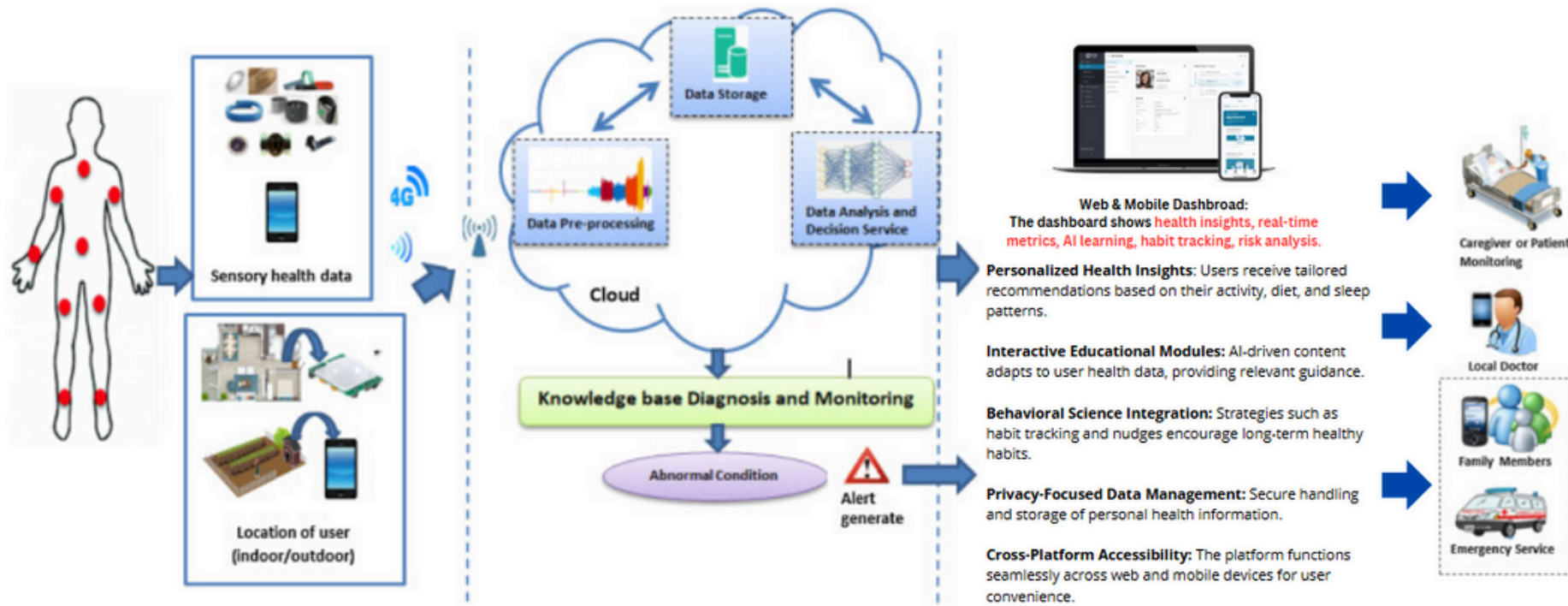


PROPOSED SOLUTION

Innovative Solution

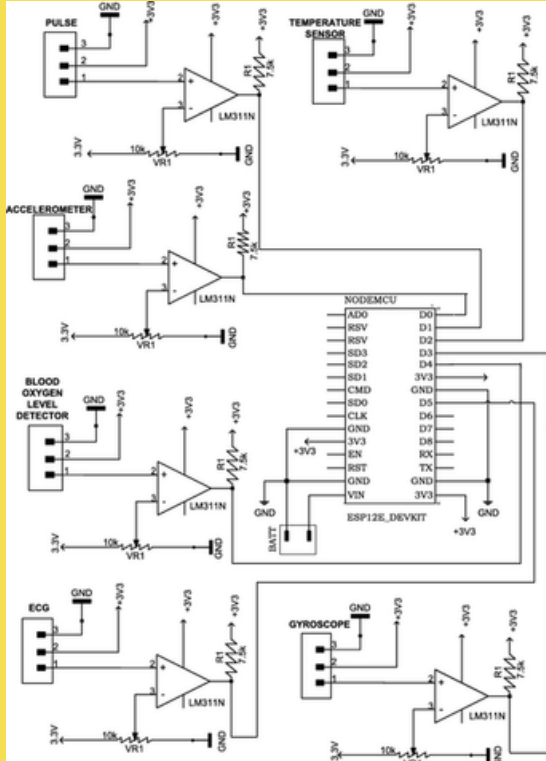


Workflow:



SmartHealth Band

The aim is to develop a smart HealthBand that continuously monitors health parameters and transmits data to the cloud. The server will process and analyze the data, providing tailored health recommendations based on activity, diet, and sleep patterns. Additionally, AI-driven educational modules will adapt to user health data, offering personalized guidance for improved well-being.



TECH STACK



1. Data Collection (Edge Layer)

Wearable sensors (Heart rate, Sleep, Activity, Diet).

IoT Gateway (Raspberry Pi) transmits data via Wi-Fi/Bluetooth/LTE.

2. Cloud Storage & Transmission

MQTT/HTTPS for secure data transfer.

NoSQL (MongoDB) & SQL (MySQL) for data.

3. AI-ML Processing & Analytics

Real-time data cleaning (Kafka & Spark).

AI models (TensorFlow & PyTorch) for behavior & risk analysis.

Personalized health recommendations via smart habits tracking.

4. User Interaction (Apps & API Layer)

Mobile & Web Apps (React, Flutter) for dashboards & tracking.

API Gateway (FastAPI & Node.js) for seamless data access.

Gamification (Streaks, Rewards, Notifications) for engagement.

5. Security & Compliance

OAuth 2.0 for authentication.

GDPR-compliant encryption (AES-256) for privacy.

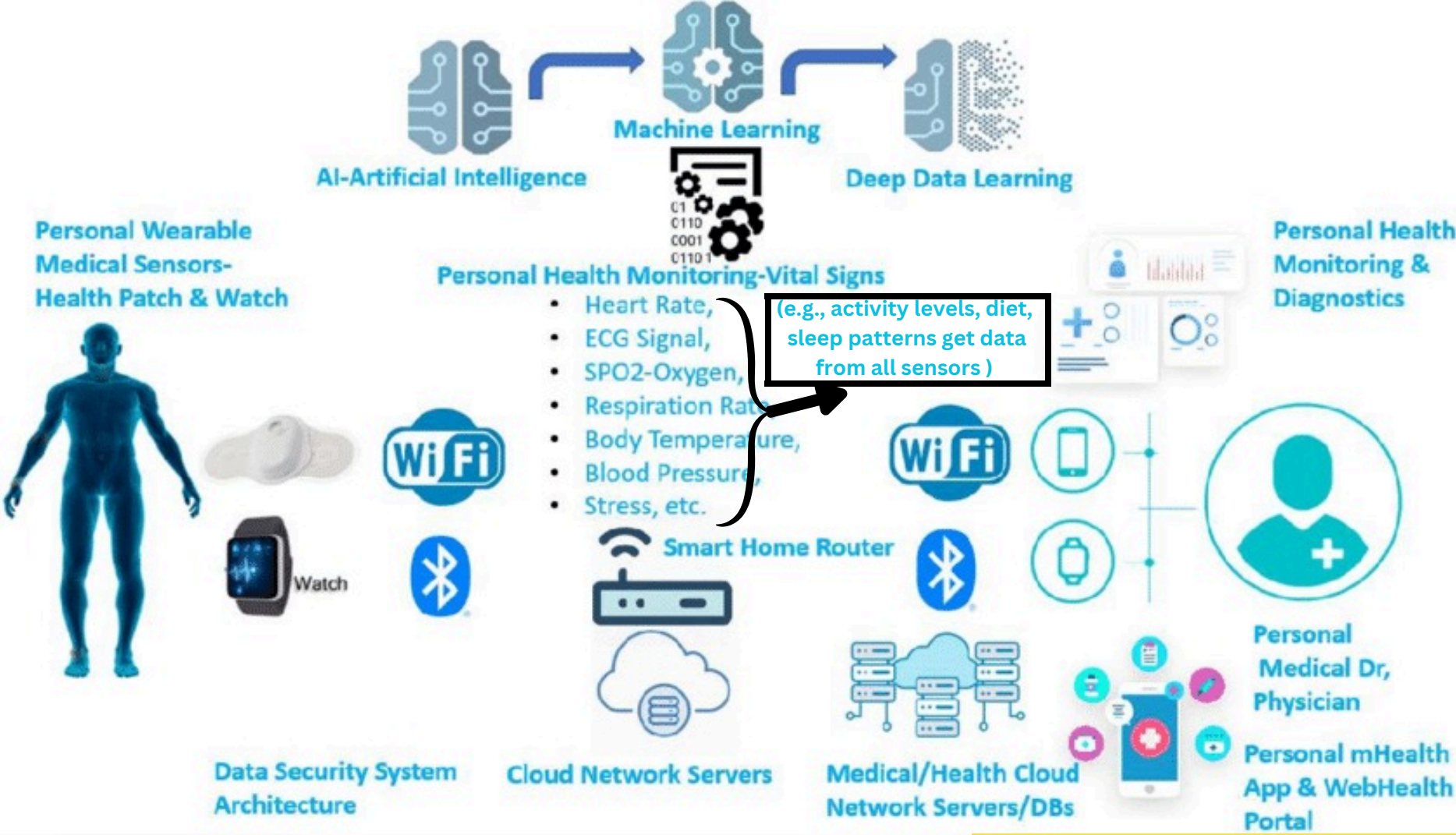
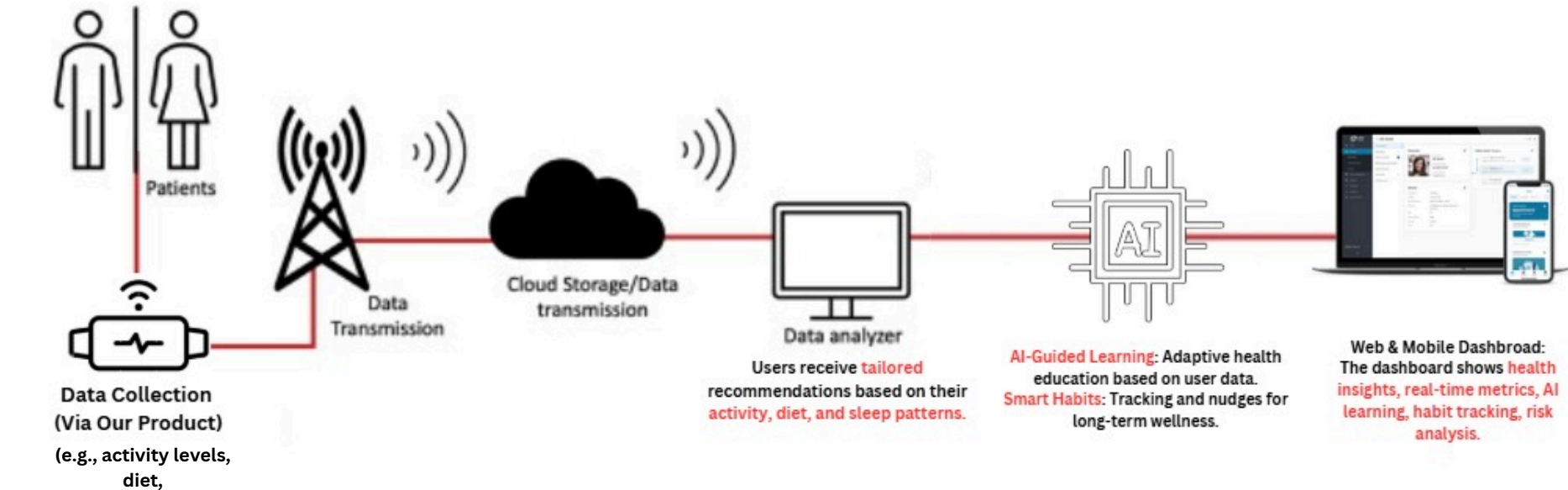
Data anonymization & federated learning for secure AI processing.

End-to-End Flow:

Sensors → IoT Gateway → Cloud → AI Processing → Insights to Users

Hardware components

Sensor Name	Type	Sensor Name	Type	Sensor Name	Type
Raspberry Pi 4	Microcontroller	MAX30205	Temp. Sensor	ADXL345	Actigraphy Sensor
Max30102	PPG Sensor	ADS1292R	ECG Sensor	Pneumotrace	Respiration Sensor
MPU6050	Gyroscope	GSR Sensor	GSR Sensor	Libre Freestyle	CGM Sensor
SGP30	Air Quality Sen.	NeuroSky TGAM	EEG Sensor	DFRobot Gravity TDS	Hydration Sensor
ML8511	UV Sensor	TSL2561	Blue Light Sen.	LoRa Module (SX1276)	Comm. Module



WHAT MAKES OUR PRODUCT DIFFERENT FROM OTHER BRANDS



Continuous Glucose Monitor (CGM) Integration

Function: Uses electrochemical sensors to monitor glucose levels in real-time via a mobile app.

Advantages:

- Prevents DKA, hypoglycemia, and hyperglycemia
- Enables better insulin dosing
- Supports insulin pump integration
- Reduces risk of heart, kidney, and nerve damage



Auto Injector Integration

Function: Spring-loaded, one-button mechanism for emergency drug delivery via the wearable band.

Advantages:

- Emergency Use: EpiPen (allergies), Diazepam (seizures), Naloxone (opioid overdose)
- Smart Injectors: Bluetooth/NFC-enabled, tracks usage, sends reminders, AI adjusts dose
- Portable and easy to use during emergencies



NFC (Near Field Communication)

Functions:

- Displays medical history for emergency treatment

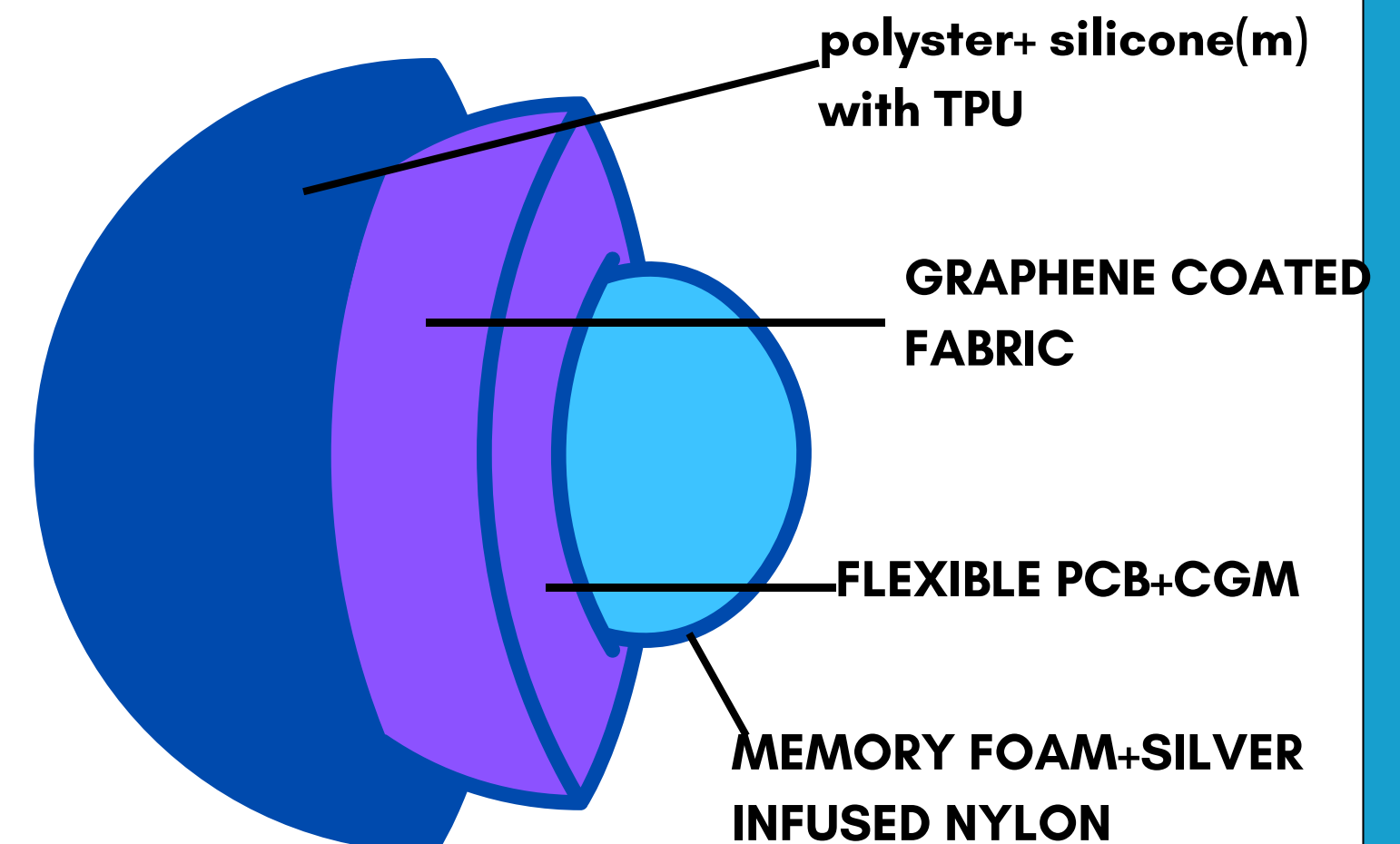
- Alerts family members during emergencies

- Detects if the band is worn; activates low-power mode if not

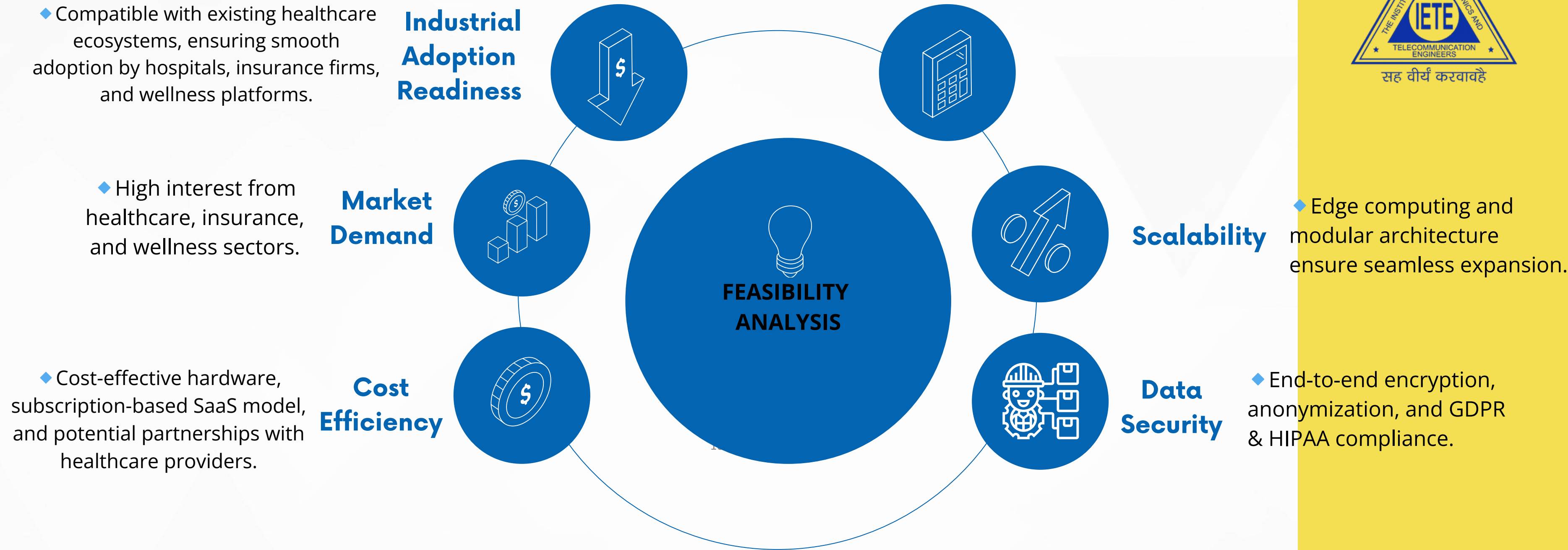
MATERIAL LAYOUT

THE DIFFERENT MATERIALS USED ARE GIVEN WITH ITS CORRESPONDING USES:

- **Outer Layer**–polyster+medical grade silicon with TPU(THERMOPLASTIC POLYURETHANE) for wear and tear,waterproffing,dust and moisture
- **Middle Layer**–graphene coated fabric for better heat dissipation $\sim 5300 \text{ W/m}\cdot\text{K}$
- **Flexible PCB+CGM**
- **INNER CORE**–memory foam with a silver infused nylon for anti bacterial properties and sweat absorption



Feasibility Analysis



The **HealthBand**, priced at **₹18,000**, stands out as one of the **most portable and advanced health monitoring devices** in the market today. With **AI-driven analytics, real-time cloud connectivity, and high-precision sensors**, it offers a **lightweight** yet powerful solution for **personalized health tracking**. Its compact design, cross-platform accessibility, and future **scalability make it a cost-effective and cutting-edge choice for users seeking real-time health insights**.

TARGET MARKET

CORE MARKET

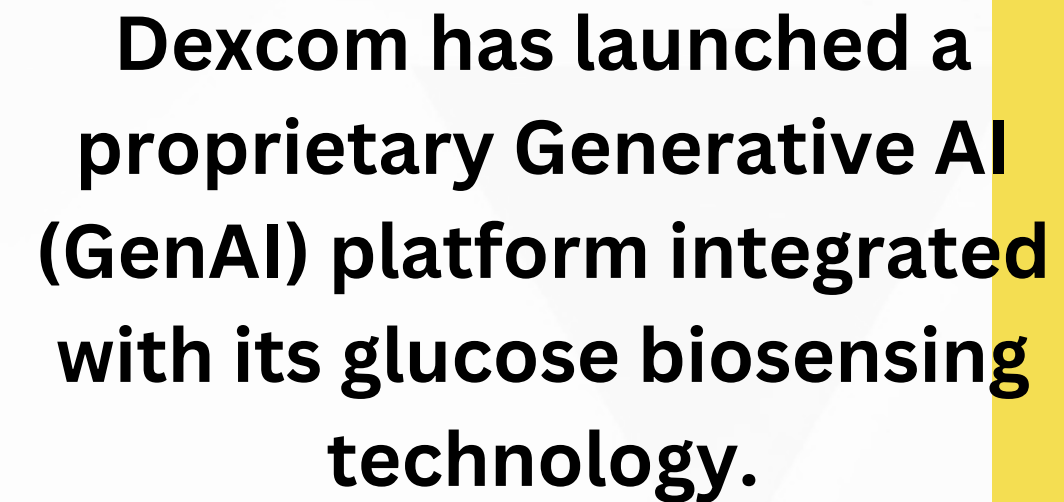
- PATIENTS WHO REQUIRE MEDICAL ATTENTION 24/7
- PEOPLE AT RISK OF OF SUDDEN HEALTH EMERGENCY
- INDIVIDUALS SUCH AS ELDERLY PEOPLE, OWEWEIGHT ,KIDS AND PREGNANT WOMEN

GROWING SEGMENT

- HEALTH-CONSCIOUS CONSUMERS
- FITNESS ENTHUSIASTS, ATHLETES, AND LONGEVITY-FOCUSED INDIVIDUALS
- USERS OF WEARABLE HEALTH TECH LIKE APPLE WATCH, FITBIT, WHOOP, OURA RING

MASS MARKET EXPANSION

- PEOPLE LOOKING FOR NUTRITION GUIDANCE
- INDIVIDUALS STRUGGLING WITH INSULIN OR OTHER REGULAR MEDICATION

[illegible]

COST ESTIMATION

KEY PCB RETAILERS

SHOGINI TECHNOARTS PVT

ANAND ELECTRONICS

VISHAL INTERNATIONAL

LIONCIRCUITS

VALUE PROPOSITIONS(PER PIECE)

COST OF THE FLEXIBLE CIRCUIT BOARD	₹3300.00
HARDWARE MEDICAL GRADE SENSORS	₹8450.00
CLOUD STORAGE(GOOGLE CLOUD)	₹200.00
SHIPPING AND SALES	₹1500.00
SERVICE AND WARRANTY	₹1000.00
FABRICATION	₹3300.00
TOTAL COST	₹17,750.00

AVAILABLE CHANNEL(AND THEIR COST IN %)

DIGITAL MARKETING	(+10%)
TRADITIONAL MAKETING	(+34%)
SOCIAL MEDIA MARKETING	(+23%)
REFERENCE MARKETING	(+28%)

COST STRUCTURE

COST OF THE MODEL	₹17,750
VALUE PREPOSITION OF THE FINISHED PRODUCT	₹17,000(PER PEICE)
FINISHED PRODUCT(COMMERCIAL AND MASS PRODUCTION)	₹16,241(8.5%LESS)



TI-BAND

FeaturesDashboardRecomendationTaskAboutContact

🍽️ Weekly Food Plan (Healthy Eating)

This meal plan supports energy levels, gut health, and better sleep.

🍳 Breakfast (6:30 AM)

Monday: Oats + Almonds + Greek Yogurt

Tuesday: Scrambled Eggs + Avocado Toast

Wednesday: Smoothie (Banana + Berries + Protein)

Thursday: Poha / Idli with Coconut Chutney

Friday: Whole Wheat Pancakes + Peanut Butter

Saturday: Sprouts & Chia

🍲 Lunch (12:00 PM)

Monday: Brown Rice + Grilled Chicken + Salad

Tuesday: Quinoa + Paneer Curry + Raita

Wednesday: Chapati + Dal + Vegetables

Thursday: Millet Bowl + Boiled Eggs + Greens

Friday: Whole Wheat Pasta + Veggies

Saturday: Dal Khichdi + Curd

🍵 Evening Snack (6:00 PM)

Monday: Nuts & Green Tea

Tuesday: Fruit + Yogurt

Wednesday: Roasted Chickpeas

Thursday: Dark Chocolate + Herbal Tea

Friday: Sweet Potato Chaat

Saturday: Mixed Seeds & Dry Fruits

Sunday: Soup + Whole Wheat Crackers

🌙 Dinner (8:00 PM)

Monday: Lentil Soup + Grilled Vegetables

Tuesday: Paneer Bhurji + Roti

Wednesday: Grilled Fish + Steamed Veggies

Thursday: Chickpea Salad + Yogurt

Friday: Moong Dal Khichdi + Buttermilk

Saturday: Light Dosa + Sambar

Sunday: Vegetable Stir-Fry +

TI-BAND

FeaturesDashboardRecomendationTaskAboutContact

🌡️ Body Temperature

36.9°C

Normal

Past 24hr: +0.2

💓 Heart Rate

73 bpm

Normal

Past 24hr: Stable

🔥 Blood Pressure

117/76

Normal

Past 24hr: -1.0

View History

🩸 Oxygen Saturation

98%

Normal

Past 24hr: +1.0

👣 Steps

Default

keep walking

Past 24hr: Avg=0

📈 Blood Glucose

Default

Normal

Past 24hr: -5 mg/dl

🔥 CALORIES

Default

Normal

Past 30d: 0

📊 ECG

Normal

Sinus rhythm

Last check: 2h ago

TI-BAND

FeaturesDashboardRecomendationTaskAboutCont

📅 Daily Schedule

Time	Activity	Notes
3:00 AM	Wake up & Hydrate	Drink 500 ml of warm water to activate digestion.
3:30 AM - 4:00 AM	Meditation & Deep Breathing	Helps wake up your brain & improve focus.
4:00 AM - 4:30 AM	Light Workout (Stretching/Yoga)	Keep the body active without over-exertion.
4:30 AM - 5:00 AM	Morning Tea / Herbal Drink	Green Tea / Lemon Honey Water / Turmeric Tea
5:00 AM - 6:00 AM	Work / Study / Productivity	Best time for deep focus.
6:00 AM - 7:00 AM	Healthy Breakfast	Protein + Good Fats + Fiber (See meal plan below)
7:30 AM - 12:00 PM	Work / Meetings / Classes	Take small breaks every 2 hours.
12:00 PM - 12:30 PM	Lunch	Balanced meal with carbs, proteins, veggies.

Predictions (Using AI & Real-Time Data)

--- Health Assessment Result ---

Predicted Health Condition: Unhealthy

--- Health Risks Detected ---

- Low Oxygen Saturation: Potential Respiratory Concerns

- Irregular Sleep Pattern: May Affect Overall Health

- Low Physical Activity: Recommend Increased Exercise

--- Recommendations ---

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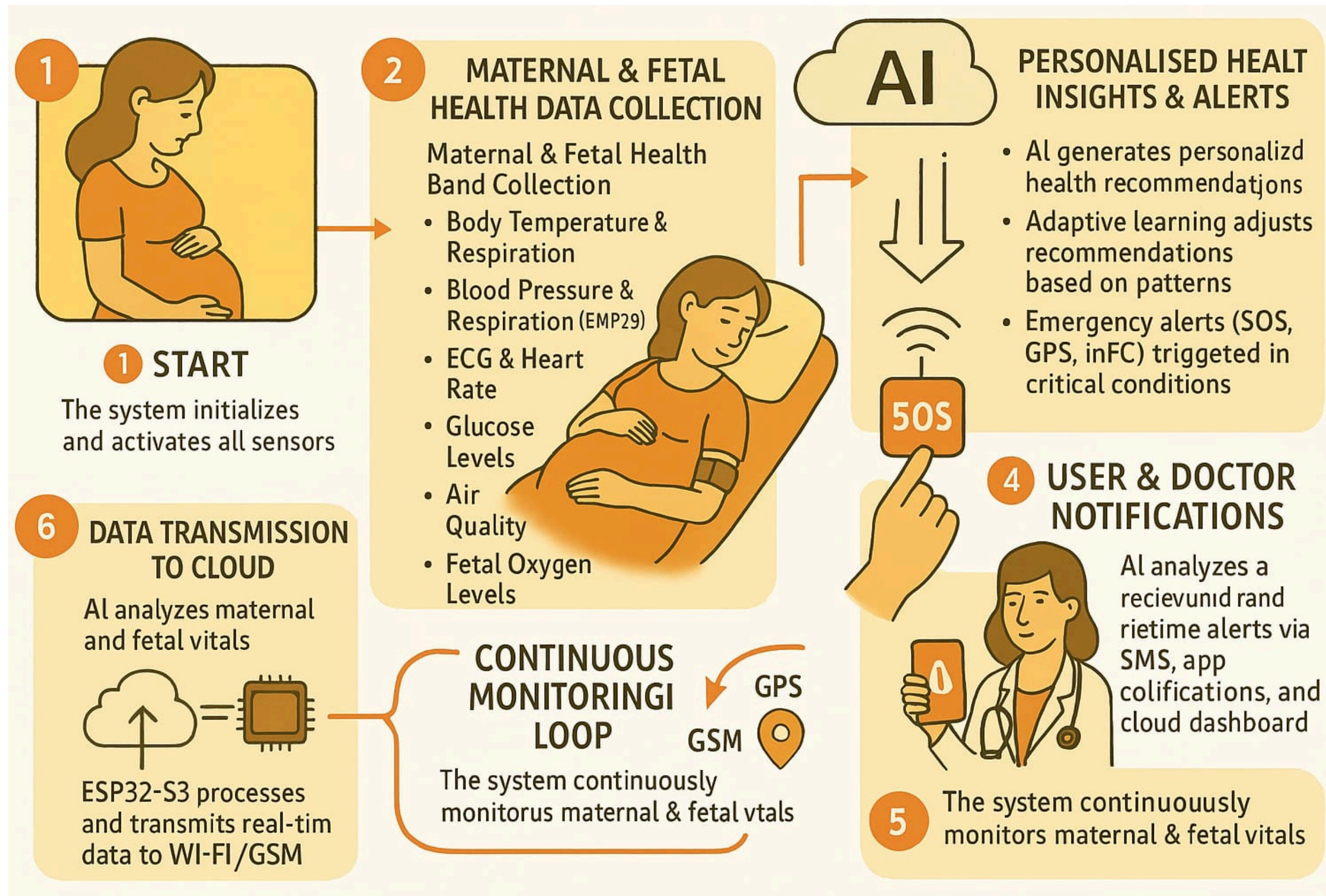
- Consider lifestyle and dietary modifications

- Risk Management: Low Oxygen Saturation: Potential Respiratory Concerns

- Risk Management: Irregular Sleep Pattern: May Affect Overall Health

- Risk Management: Low Physical Activity: Recommend Increased Exercise

IN CASE OF THE BAND IS USED FOR THE MATERNAL WOMEN:



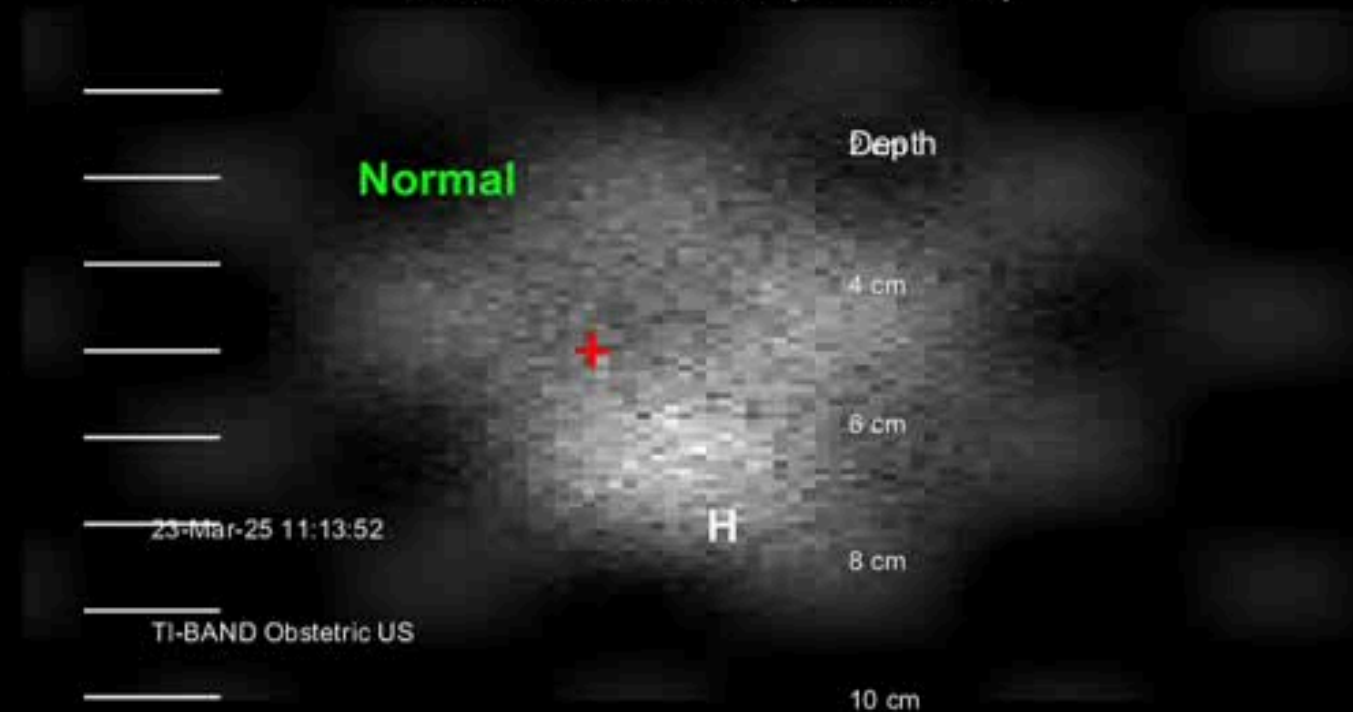
The flowchart represents an AI-powered maternal and fetal health monitoring system that continuously tracks vital health parameters. The process begins with the system initialization, activating sensors that measure body temperature, respiration, blood pressure, ECG, blood oxygen levels, fetal kicks, and air quality. AI analyzes this data to generate personalized health recommendations and detect anomalies. If an emergency is detected, the system triggers alerts via SOS, GPS, and NFC to ensure immediate attention.

Once an alert is generated, real-time notifications are sent to doctors and caregivers through SMS, mobile app notifications, and cloud dashboards. The system continuously monitors health data, looping back for ongoing tracking to ensure proactive healthcare. All processed data is securely transmitted to the cloud via Wi-Fi or GSM, allowing remote access for medical professionals and enabling timely interventions when necessary.





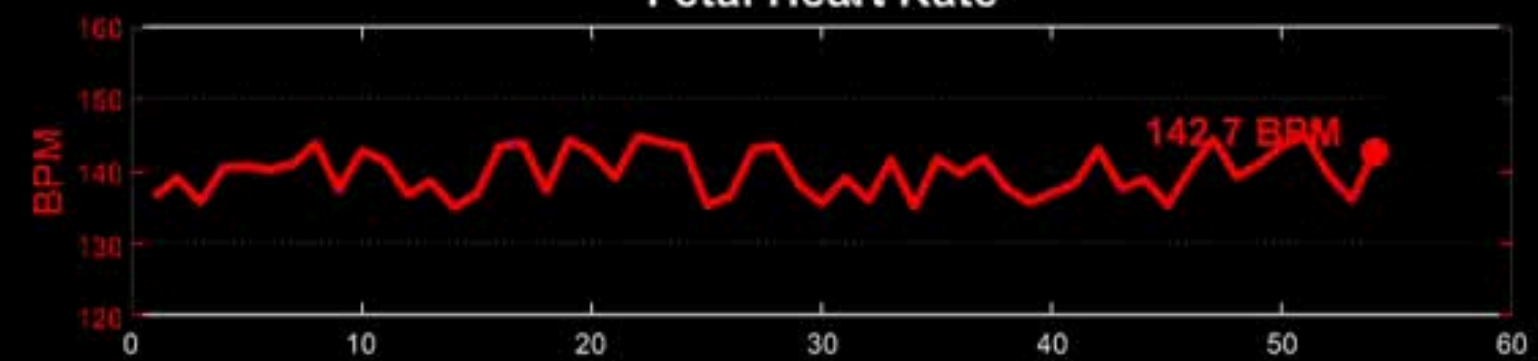
Fetal Ultrasound (28 weeks)



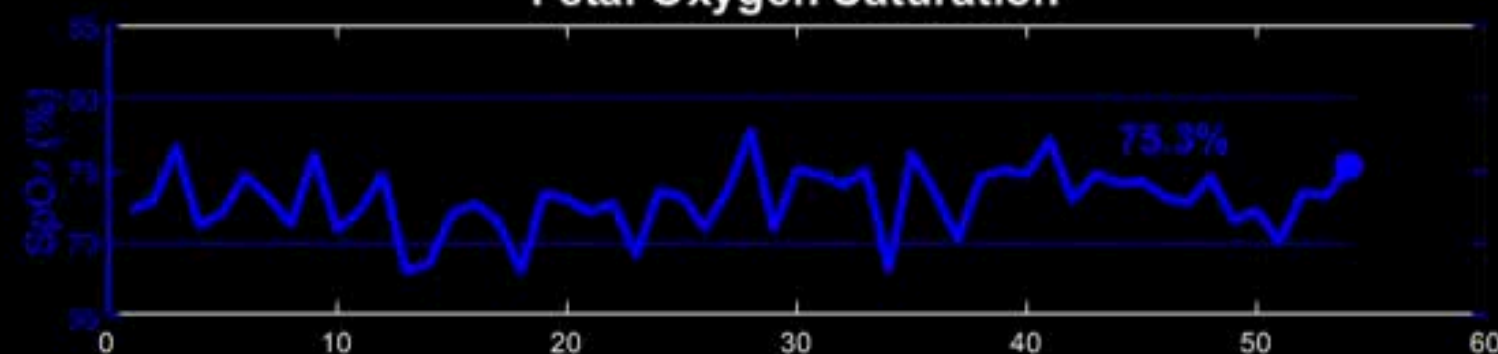
Fetal Movement Activity



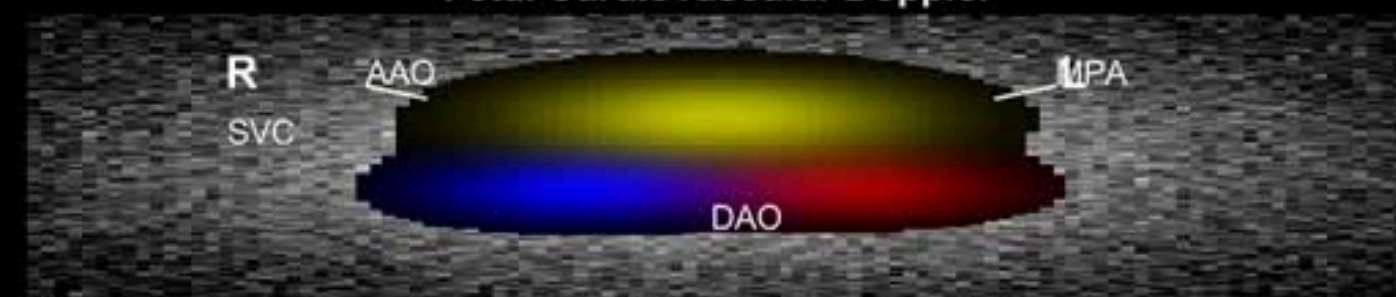
Fetal Heart Rate



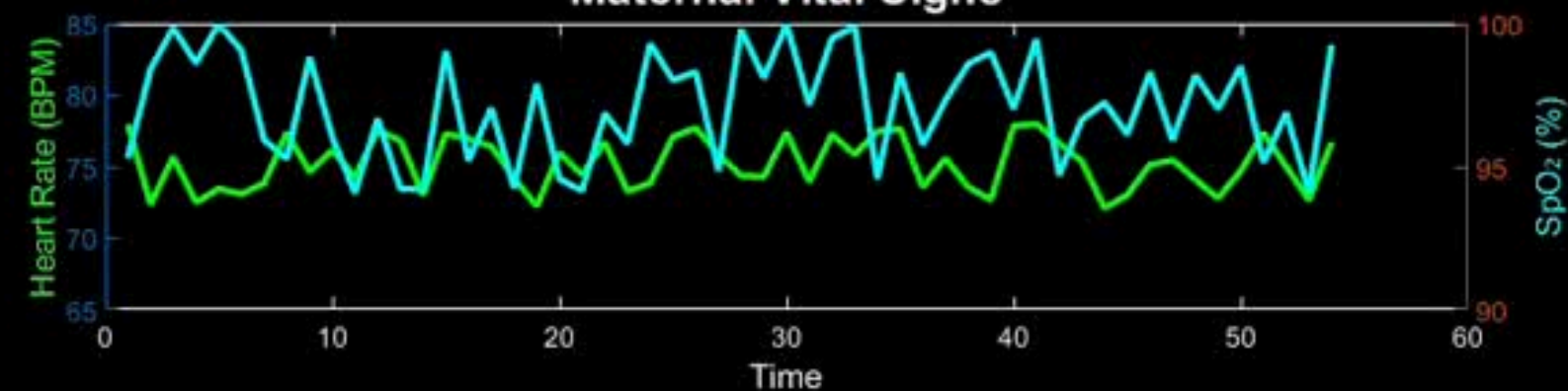
Fetal Oxygen Saturation



Fetal Cardiovascular Doppler



Maternal Vital Signs



Fetal Assessment Summary

CURRENT STATUS:

Movement: normal
Head Position: upward
Kicks Detected: 0
Activity Level: LOW

FETAL VITALS:

HR: 142.7 BPM
SpO2: 75.3%

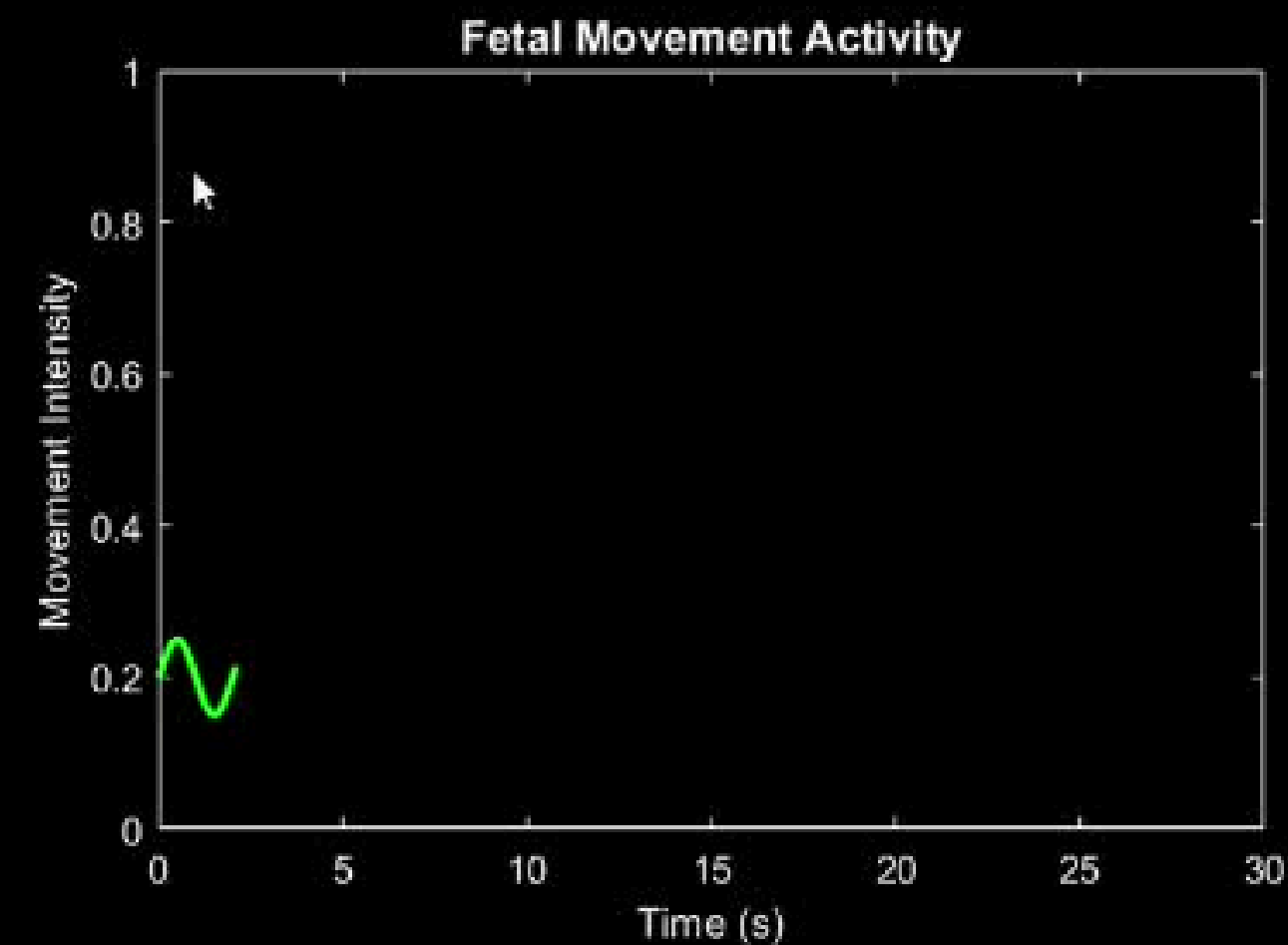
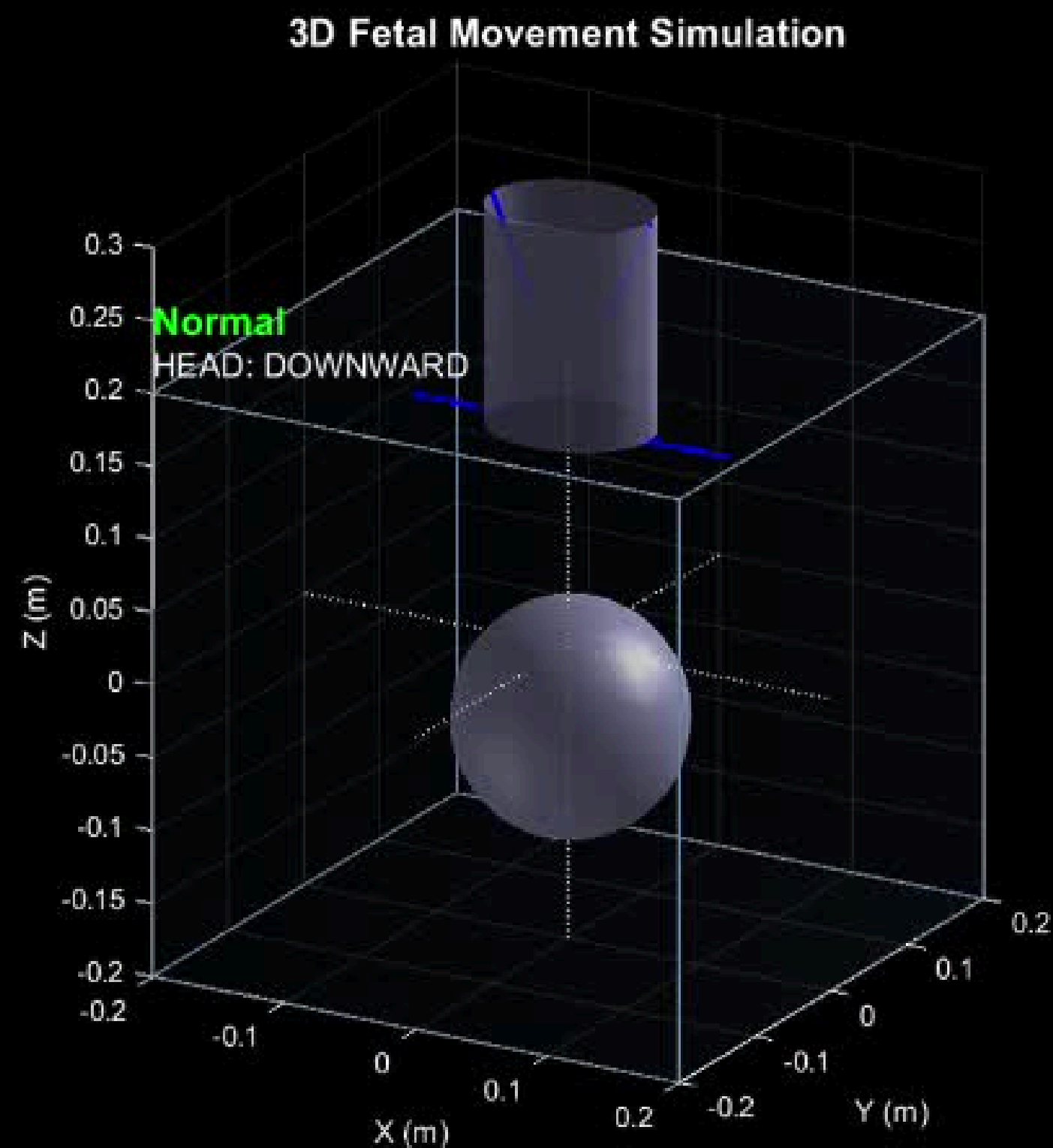
FETAL ASSESSMENT:

Gestational Age: 28 weeks
Biophysical Profile: 6/10
Est. Fetal Weight: 2300 g
Presentation: Breech

SESSION INFO:

Session time: 5.3 s
Data points: 54

23-Mar-25 11:13:52



Fetal Movement Monitor

CURRENT STATUS: MOVEMENT STATS:

Movement: normal **Kicks: 0**

Head Position: downward **Activity: LOW**

POSITION DATA: Session time: 2.1 s

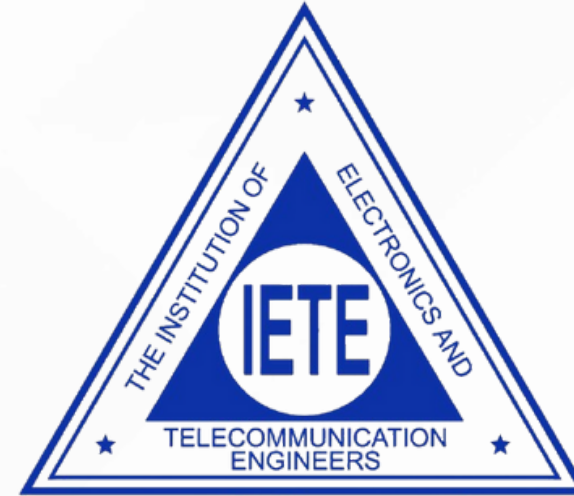
X: 0.001 m Data points: 22

Y: 0.002 m

Z: 0.101 m

22-Mar-25 22:39:11

Data From the TI BAND



सह वीर्यं करवावहे

THANK YOU

<Team Code>